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United States Patent	12412210
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Huber; Theresa et al.

Consolidation of data sources for expedited validation of risk assessment data

Abstract

Systems and methods for consolidating data sources for expediting validation processes are described herein. A user interface can be provided to a first entity. First risk assessment data associated with the first entity can be received via the user interface. Second risk assessment data associated with the first entity can be received. The first risk assessment data and the second risk assessment data can be validated. The first risk assessment data and the second risk assessment data can be output for display on the user interface. A responsive message including the validated first risk assessment data and the validated second risk assessment data can be transmitted to a remote computing device for use in controlling access of the first entity to one or more interactive computing environments.

Inventors: Huber; Theresa (Ballwin, MO), John; Dileep (St. Louis, MO), Mudduluri; Balaji (Chesterfield, MO), Arshad; Muhammad (Valley Park, MO), Katabathuni; Kishore Kumar (Chesterfield, MO), Golla; Saratchandra (O'Fallon, MO), Vontari; Akshay Reddy (Irving, TX)

Applicant: EQUIFAX INC. (Atlanta, GA)

Family ID: 86768564

Assignee: EQUIFAX INC. (Atlanta, GA)

Appl. No.: 18/067578

Filed: December 16, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230196455 A1	Jun. 22, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: **G06Q40/03** (20230101)

U.S. Cl.:

CPC **G06Q40/03** (20230101);

Field of Classification Search

USPC: None

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Primary Examiner: Nolan; Peter D

Assistant Examiner: Cobb; Matthew

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to provisional application No. 63/265,609 filed Dec. 17, 2021, entitled “CONSOLIDATION OF DATA SOURCES FOR EXPEDITED VALIDATION OF RISK ASSESSMENT DATA,” the entirety of which is incorporated by reference herein for all purposes.

TECHNICAL FIELD

(1) The present disclosure relates generally to risk assessment. More specifically, but not by way of limitation, this disclosure relates to consolidating data sources for expediting validation of risk assessment data.

BACKGROUND

(2) Risk assessment can be performed by various computing devices. In some examples, risk assessment can involve analyzing data to determine a risk value. A computing device may be configured to analyze data and determine risk based on a single source of data or using a single process. But, in some cases, the single source of data may not provide accurate risk assessment input. Additionally, the computing device may not be configured to analyze more than the single source of data or execute more than a single process for performing the risk assessment.

SUMMARY

- (3) Various aspects of the present disclosure provide systems and methods for consolidating data sources for expediting validation processes. A user interface can be provided to a first entity. The user interface can include user interface elements corresponding to risk assessment indicators. First risk assessment data associated with the first entity can be received. Second risk assessment data associated with the first entity can be received from a second entity. The first risk assessment data and the second risk assessment data can be validated. The first risk assessment data and the second risk assessment data can be output for display on the user interface. A responsive message including at least the validated first risk assessment data and the validated second risk assessment data can be transmitted to a remote computing device for use in controlling access of the first entity to one or more interactive computing environments.
- (4) In other aspects, a system can include a processor and a non-transitory computer-readable medium that can include instructions that are executable by the processor to cause the processor to perform various operations. The system can provide a user interface to a first entity. The user interface can include user interface elements corresponding to risk assessment indicators. The system can receive, via the user interface, first risk assessment data that is associated with the first entity. The system can receive, based on the first risk assessment data and from a second entity, second risk assessment data that is associated with the first entity. The system can validate the first risk assessment data and the second risk assessment data. The system can output the first risk assessment data and the second risk assessment data for display on the user interface. The system can transmit, to a remote computing device, a responsive message including at least the validated first risk assessment data and the validated second risk assessment data for use in controlling access of the first entity to one or more interactive computing environments.
- (5) In other aspects, a non-transitory computer-readable medium can include instructions that are executable by a processing device for causing the processing device to perform various operations. The operations can include providing a user interface to a first entity. The user interface can include user interface elements corresponding to risk assessment indicators. The operations can include receiving, via the user interface, first risk assessment data that is associated with the first entity. The operations can include receiving, based on the first risk assessment data and from a second entity, second risk assessment data that is associated with the first entity. The operations can include validating the first risk assessment data and the second risk assessment data. The operations can include outputting the first risk assessment data and the second risk assessment data for display on the user interface. The operations can include transmitting, to a remote computing device, a responsive message including at least the validated first risk assessment data and the validated second risk assessment data for use in controlling access of the first entity to one or more interactive computing environments.
- (6) This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification, any or all drawings, and each claim.
- (7) The foregoing, together with other features and examples, will become more apparent upon referring to the following specification, claims, and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram depicting an example of an operating environment that includes a risk assessment computing system that can consolidate data sources for expediting validation of risk assessment data according to certain aspects of the present disclosure.
- (2) FIG. 2 is a flow chart depicting an example of a process for consolidating data sources for

expediting risk assessment validation according to certain aspects of the present disclosure.

(3) FIG. 3 is a flow diagram depicting a flow of risk assessment data in a network environment for consolidating data sources to expedite validation of the risk assessment data according to certain aspects of the present disclosure.

(4) FIG. 4 shows an example of a user interface that can be used to consolidate data sources for expediting validation of risk assessment data according to certain aspects of the present disclosure.

(5) FIG. 5 shows an example of a subsequent user interface that can be used to receive risk assessment data to consolidate data sources for expediting validation of risk assessment data according to certain aspects of the present disclosure.

(6) FIG. 6 is a block diagram depicting an example of a computing system suitable for implementing certain aspects of the present disclosure.

DETAILED DESCRIPTION

(7) Various techniques using computing systems can be used to output various information. For example, a computing device or system can predict risk assessment values (e.g., risk scores associated with granting access to an online computing environment, or credit scores, etc.) with respect to one or more entities that can be used for making decisions about the entities. But, the risk assessment values may involve one or more validation processes. For example, a first entity (e.g., a user, a consumer, or other individual) may submit a first set of data (e.g., system configurations and computing resource usage for a computing system, asset, credit for an individual, or other suitable data) for validation, and a second entity (e.g., a host of the online computing environment, a financial institution or other suitable entity) can submit a second set of data (e.g., system configurations and computing resource usage of the computing system, assets financial records of the individual, etc.) for validation. Other computing systems may not be configured to validate both sets of data, or any additional received data, and may wait or otherwise delay the validation process until receiving validation of all data from separate computing systems. While this example is described with respect to two datasets, any suitable number of datasets, such as less than two or more than two, can be submitted for validation, and the other computing systems may not be able to validate the received data.

(8) Certain aspects and features of the present disclosure that consolidate data sources for expediting validation processes with respect to risk assessment data can address one or more issues identified above. For example, a computing system can receive a set of risk assessment data from various sources and can validate the risk assessment data. The set of risk assessment data can include more than one risk assessment indicator or value, and the various sources can include an individual entity, a service provider, a government institution, or other suitable sources of the data. A risk assessment indicator or value can include a risk score, an asset ratio, a listing of assets, government records or information, other suitable risk assessment indicators, or a combination thereof. By consolidating the risk assessment data and validating the risk assessment data before initiating an interaction, such as requesting access to computational resources, etc., the computing system can change its configuration or computing resource usage, such as decreasing the amount of memory, processing power, network usage, or other resources used for the interaction and can make the interaction and validation process faster and more efficient.

(9) In some examples, the computing system can provide a user interface for an entity. The user interface can include a set of user interface elements that can correspond to various risk assessment indicators. The user interface can include other suitable user interface elements such as user interface elements for allowing the entity to provide goals and other suitable information. A first risk assessment indicator can include a risk score, such as a credit score, a second risk assessment indicator can include an asset ratio, such as a debt-to-income ratio, and a third risk assessment indicator can include an amount of assets for completing an interaction, for example an amount of resources requested for granting access to an interactive computing environment, or the like. Other amounts and other types of suitable risk assessment indicators can be presented or otherwise

displayed by the user interface. The user interface can include other suitable elements, information, etc., for display to the first entity or to other suitable entities.

(10) The entity can interact with each user interface element for causing the computing system to receive risk assessment data. For example, the entity can interact with a first user interface element to provide risk assessment data for the first risk assessment indicator. In response to the entity interacting with the first user interface element, the computing system can provide a subsequent user interface to guide the entity to provide accurate and complete information for determining the risk assessment indicator. The entity can similarly interact with each user interface element for providing the computing system with accurate and complete information for determining each risk assessment indicator.

(11) In some examples, interacting with one or more of the user interface elements can cause the computing system to communicate with separate computing systems. For example, the entity can interact with the user interface element corresponding to the third risk assessment indicator, and the computing system may provide a subsequent user interface to the entity that can allow the entity to login to a service-provider institution such as a bank, a data provider, or the like. Upon the entity logging into the service-provider institution, a token can be generated by the computing system that can be used to receive information from the service-provider institution. For example, the computing system can make an application programming interface (API) call to the service-provider institution and can use the token to receive data relating to the entity from the financial institution. The data can include an asset report or other suitable information associated with the entity and usable for determining the third risk assessment indicator. In some examples, the asset report can include a predefined format, and the computing system can extract information from the asset report that can be used to determine the third risk assessment indicator.

(12) As described herein, certain aspects provide improvements to controlling access to computing resources. For example, validated risk assessment data can be used by a risk assessment system to more accurately and efficiently control access to computing resources such as an interactive computing environment that can provide computing resources, such as computational processing power, computer memory, storage resources, network resources, and the like, to the entity. In particular, data consolidation techniques can be used by the risk assessment system to more accurately (e.g., compared to systems using other techniques), and using fewer resources, validate risk assessment data. Based on the validated risk assessment data, the risk assessment system can more accurately and efficiently control access to the computing resources. Compared with other computational techniques, consolidating risk assessment data requires fewer computing resources (e.g., CPU times) and storage resource (e.g., memory usage) to validate risk assessment data. Further, the risk assessment data valuation presented herein can also reduce the amount of time used for the validation by consolidating data received from multiple resources and performing the validation without waiting for all the data becoming available.

(13) These illustrative examples are given to introduce the reader to the general subject matter discussed here and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the illustrative examples but, like the illustrative examples, should not be used to limit the present disclosure.

(14) Operating Environment Example for Validation of Consolidated Risk Assessment Data

(15) Referring now to the drawings, FIG. 1 is a block diagram depicting an example of an operating environment **100** that includes a risk assessment computing system **130** that can consolidate data sources for expediting validation of risk assessment data. FIG. 1 depicts examples of hardware components of a risk assessment computing system **130**, according to some aspects. The risk assessment computing system **130** can be a specialized computing system that may be used for processing large amounts of data using a large number of computer processing cycles. The

risk assessment computing system **130** can include a risk assessment server **118** for validating risk assessment data from various sources. In some examples, the risk assessment computing system **130** can include other suitable components, servers, subsystems, etc.

(16) The risk assessment server **118** can include one or more processing devices that can execute program code, such as a risk assessment mobile application **114**, which can be used by a first entity such as a consumer or other suitable individual, or a risk assessment web application **115**, which can be used by a second entity such as a provider of financial products or other suitable services. The program code can be stored on a non-transitory computer-readable medium or other suitable medium. The risk assessment mobile application **114** can execute one or more processes to present a user interface for receiving risk assessment data. The risk assessment server **118** can then perform risk assessment validation operations for validating the received risk assessment data, for example using other suitable modules, models, components, etc. of the risk assessment server **118**. In some aspects, the risk assessment mobile application **114** can validate risk assessment data received from an entity and by utilizing pre-existing risk assessment data **124**, which can include historical risk assessment data or other suitable data. The pre-existing risk assessment data **124** can be stored in one or more network-attached storage units on which various repositories, databases, or other structures are stored. Examples of these data structures can include the risk data repository **122**.

(17) Network-attached storage units may store a variety of different types of data organized in a variety of different ways and from a variety of different sources. For example, the network-attached storage unit may include storage other than primary storage located within the risk assessment server **118** that is directly accessible by processors located therein. In some aspects, the network-attached storage unit may include secondary, tertiary, or auxiliary storage, such as large hard drives, servers, and virtual memory, among other types of suitable storage. Storage devices may include portable or non-portable storage devices, optical storage devices, and various other mediums capable of storing and containing data. A machine-readable storage medium or computer-readable storage medium may include a non-transitory medium in which data can be stored and that does not include carrier waves or transitory electronic signals. Examples of a non-transitory medium may include, for example, a magnetic disk or tape, optical storage media such as a compact disk or digital versatile disk, flash memory, memory devices, or other suitable media.

(18) Furthermore, the risk assessment computing system **130** can communicate with various other computing systems. The other computing systems can include client computing systems **104**, entity computing systems **106**, such as smartphones, personal computers, and the like, external resource **107**, such as computing systems owned or operated by service-providers, financial institutions, and the like, and other suitable computing systems. For example, client computing systems **104** may transmit risk assessment queries, for example via the risk assessment web application **115**, to the risk assessment server **118** for risk assessment, or may transmit signals to the risk assessment server **118** that control or otherwise influence different aspects of the risk assessment computing system **130**. The risk assessment computing system **130** may also interact, for example via the risk assessment mobile application **114**, with entity computing systems **106** via one or more public data networks **108** to facilitate interactions between users of the entity computing systems **106** and interactive computing environments provided by the client computing systems **104**. In some examples, the risk assessment computing system **130** can provide a user interface to the entity computing system **106** for receiving risk assessment data from the first entity, and the risk assessment computing system **130** can transmit validated risk assessment data to the client computing system **104** for making decisions about the first entity. In some examples, the risk assessment computing system **130** can additionally communicate with the external resource **107**, for example to receive additional risk assessment data, through the public data network **108**.

(19) Each client computing system **104** may include one or more third-party devices, such as individual servers or groups of servers operating in a distributed manner. A client computing system **104** can include any computing device or group of computing devices operated by a service

provider, such as a seller, lender, or other suitable provider of products or services. The client computing system **104** can include one or more server devices. The one or more server devices can include or can otherwise access one or more non-transitory computer-readable media.

(20) The client computing system **104** can further include one or more processing devices that can provide an interactive computing environment, such as a user interface, to perform one or more of the operations described herein. The interactive computing environment can include executable instructions stored in one or more non-transitory computer-readable media. The instructions providing the interactive computing environment can configure one or more processing devices to perform one or more of the operations described herein. In some aspects, the executable instructions for the interactive computing environment can include instructions that provide one or more graphical interfaces. The graphical interfaces can be used by an entity computing system **106** to access various functions of the interactive computing environment. For instance, the interactive computing environment may transmit data to and receive data, for example via the graphical interface, from an entity computing system **106** to shift between different states of the interactive computing environment, where the different states allow one or more electronics interactions between the entity computing system **106** and the risk assessment computing system **130** to be performed.

(21) In some examples, the client computing system **104** may include other computing resources associated therewith, for example not shown in FIG. **1**, such as server computers hosting and managing virtual machine instances for providing cloud computing services, server computers hosting and managing online storage resources for users, server computers for providing database services, and others. The interaction between the entity computing system **106** and the risk assessment computing system **130** may be performed through graphical user interfaces, such as the user interface, that can be presented by the risk assessment computing system **130** to the entity computing system **106**, or through application programming interface (API) calls or web service calls, or through other suitable network communication techniques.

(22) An entity computing system **106** can include any computing device or other communication device operated by a user or entity, such as a consumer or other suitable individual. The entity computing system **106** can include one or more computing devices, such as laptops, smartphones, and other personal computing devices. A entity computing system **106** can include executable instructions stored in one or more non-transitory computer-readable media. The entity computing system **106** can also include one or more processing devices that can execute program code to perform one or more of the operations described herein. In various examples, the entity computing system **106** can allow a user to access certain online services from a client computing system **104** or other computing resources, to engage in mobile commerce with a client computing system **104**, to obtain controlled access to electronic content hosted by the client computing system **104**, and the like.

(23) The entity can use the entity computing system **106** to engage in an electronic interaction with a client computing system **104** via an interactive computing environment. The risk assessment computing system **130** can receive a request, for example from the client computing system **104**, to determine risk associated with the entity and can use risk assessment values to determine and output the risk. An electronic interaction between the entity computing system **106** and the client computing system **104** can include, for example, the entity computing system **106** being used to request a financial loan, online computing resources, or other suitable services or products from the client computing system **104**, and so on. An electronic interaction between the entity computing system **106** and the client computing system **104** can also include, for example, one or more queries for a set of sensitive or other controlled data, accessing online financial services provided via the interactive computing environment, submitting an online application, for example for online computing resources, to the client computing system **104** via the interactive computing environment, operating an electronic tool within an interactive computing environment hosted by

the client computing system (e.g., a content-modification feature, an application-processing feature, etc.).

(24) In some aspects, an interactive computing environment implemented through the client computing system **104** can be used to provide access to various online functions. As a simplified example, a user interface or other interactive computing environment provided by the client computing system **104** can include electronic functions for requesting computing resources, online storage resources, network resources, database resources, or other types of resources. In another example, a website or other interactive computing environment provided by a client computing system **104** can include electronic functions for obtaining one or more financial services, such as an asset report, management tools, credit card application and transaction management workflows, electronic fund transfers, etc. An entity computing system **106** can be used to request access to the interactive computing environment provided by the client computing system **104**. The client computing system **104** can submit a request for risk assessment to the risk assessment computing system **130** and can selectively grant or deny access to various electronic functions based on the risk assessment performed by the risk assessment computing system **130**. Based on the request, the risk assessment computing system **130** can collect or otherwise receive risk assessment data associated with the first entity and communicate with the risk assessment server **118** for risk assessment. Based on the risk indicators provided and validated by the risk assessment server **118**, the client computing system **104** can determine whether to grant the access request of the entity computing system **106** to certain features of the interactive computing environment.

(25) In a simplified example, the system depicted in FIG. **1** can configure the risk assessment server **118** to be used for accurately consolidating and validating risk assessment indicators, such as risk scores, existing resources (e.g., assets), and the like. The risk assessment server **118** can receive risk assessment data from various sources, such as the client computing system **104**, the entity computing system **106**, the external resource **107**, etc., via a user interface provided to the entity computing system **106**. The risk assessment server **118** can validate the risk assessment data using pre-existing risk assessment data **124** or other suitable data and can output the validated risk assessment data for display on the entity computing system **106**, or other suitable display device, for the entity to decide whether to proceed with an interaction.

(26) In some examples, the validated risk assessment indicators can be utilized by the service provider, for example using the client computing system **104**, to determine whether the risk associated with the entity accessing a service provided by the service provider exceeds a threshold, thereby granting or denying access of the entity to an interactive computing environment implementing the service. If the service provider determines that the validated risk assessment indicators are lower than a threshold risk indicator value, then the client computing system **104** associated with the service provider can generate or otherwise provide access permission to the entity computing system **106** that requested the access. The access permission can include, for example, cryptographic keys used to generate valid access credentials or decryption keys used to decrypt access credentials. The client computing system **104** associated with the service provider can also allocate resources to the entity and provide a dedicated web address for the allocated resources to the entity computing system **106**, for example, by adding the entity computing system **106** in the access permission. With the obtained access credentials or the dedicated web address, the entity computing system **106** can establish a secure network connection to the computing environment hosted by the client computing system **104** and access the resources via invoking API calls, web service calls, HTTP requests, or other proper mechanisms.

(27) Each communication within the operating environment **100** may occur over one or more data networks, such as a public data network **108**, a network **116** such as a private data network, or some combination thereof. A data network may include one or more of a variety of different types of networks, including a wireless network, a wired network, or a combination of a wired and wireless network. Examples of suitable networks include the Internet, a personal area network, a local area

network (“LAN”), a wide area network (“WAN”), or a wireless local area network (“WLAN”). A wireless network may include a wireless interface or a combination of wireless interfaces. A wired network may include a wired interface. The wired or wireless networks may be implemented using routers, access points, bridges, gateways, or the like, to connect devices in the data network.

(28) The number of devices depicted in FIG. 1 is provided for illustrative purposes. Different numbers of devices may be used. For example, while certain devices or systems are shown as single devices in FIG. 1, multiple devices may instead be used to implement these devices or systems. Similarly, devices or systems that are shown as separate, such as the risk assessment server **118** and the risk data repository **122**, may be instead implemented in a single device or system.

(29) Techniques for Expediting Validation of Risk Assessment Data

(30) FIG. 2 is a flow chart depicting an example of a process **200** for consolidating data sources for expediting risk assessment validation according to certain aspects of the present disclosure. One or more computing devices, such as the risk assessment computing system **130**, can implement operations depicted in FIG. 2 by executing suitable program code such as the risk assessment mobile application **114**. For illustrative purposes, the process **200** is described with reference to certain examples depicted in the figures. Other implementations, however, are possible.

(31) At block **202**, the process **200** involves providing a user interface to a first entity, which can include an entity seeking access to web resources, or the like. In some examples, the user interface can be provided by the risk assessment server **118** or other suitable computing system. The user interface can include a set of user interface elements that can correspond to a set of risk assessment indicators. For example, a first user interface element can correspond to a first risk assessment indicator, and so on. The set of user interface elements can be configured to provide subsequent user interfaces in response to an interaction from the first entity. For example, a subsequent first user interface can be provided to the first entity in response to the entity interacting with the first user interface element, and so on. The user interface can include other suitable user interface elements and other suitable components for receiving risk assessment data.

(32) At block **204**, the process **200** involves receiving first risk assessment data from the first entity via the user interface. The first risk assessment data can include risk score information, such as a credit score, income information, debt information, asset information, authorization information, or other suitable risk assessment data provided by the first entity. The first entity can input the first risk assessment data into subsequent user interfaces provided by the risk assessment server **118**. For example, the first entity can interact with the first user interface element, which can cause the first subsequent user interface to be provided to the first entity. The first subsequent user interface can guide the first entity for providing accurate and complete information for determining a first risk indicator. The risk assessment server **118** can receive, for example via the first subsequent user interface, the first risk assessment data for consolidation. The first entity can interact with the remaining user interface elements, which may include the second user interface element, the third user interface element, and any remaining user interface elements, in similar fashions for providing accurate and complete first risk assessment data. Examples of the first user interface element, the second user interface element, and the third user interface element are provided below with respect to FIG. 4.

(33) In some examples, receiving the first risk assessment data can involve receiving information to pre-fill an application. The application can be a universal residential loan application (URLA, form 1003), an application for requisitioning online computing resources, or other suitable application associated with the first entity. The risk assessment server **118** can pre-fill the application using the received first risk assessment data and can perform other suitable operations with respect to the application. For example, the risk assessment server **118** can submit the application subsequent to completion and approval of the application by the first entity, and the like.

(34) At block **206**, the process **200** involves receiving second risk assessment data from a second

entity. The second risk assessment data can include risk score information, asset data, financial records, or other suitable risk assessment data. The second entity can be or otherwise include a service-providing entity, an investment firm, or other suitable entity associated with the first entity. For example, the second entity can be a financial institution with which the first entity controls an account. In some examples, the first entity can log into an account with the second entity by providing input to a subsequent user interface provided by the risk assessment server **118**. The login performed by the first entity can generate a token that can be used by the risk assessment server **118** to access the second risk assessment data. For example, the risk assessment server **118** can use the token to receive an asset report or other suitable report from the second entity. The risk assessment server **118** can parse the asset report and extract the second risk assessment data, which may include asset lists, historical bank records, and the like.

(35) At block **208**, the process **200** involves validating risk assessment data received from the first entity and the second entity. The risk assessment server **118** can validate the first risk assessment data and the second risk assessment data. For example, the risk assessment server **118** can access and receive the pre-existing risk assessment data **124**. The pre-existing risk assessment data **124** can include credit scores or other risk scores, tax returns or other government records, asset reports, employment information, identification information, or other suitable risk assessment data. The risk assessment server **118** can compare the pre-existing risk assessment data **124** to the first risk assessment data and the second risk assessment data for validating at least a subset of the first risk assessment data, at least a subset of the second risk assessment data, or a combination thereof. For example, if the risk assessment server **118** determines that the first risk assessment data and the second risk assessment data are the same as, or similar within a threshold value compared to, the pre-existing risk assessment data **124**, the risk assessment server **118** can validate the first risk assessment data and the second risk assessment data. Alternatively, if the risk assessment server **118** determines that the first risk assessment data or the second risk assessment data is not the same as or similar to the pre-existing risk assessment data **124**, then the risk assessment server **118** may not validate the first risk assessment data and the second risk assessment data. In some examples, the risk assessment server **118** can additionally use the second risk assessment data to validate the first risk assessment data. For example, the risk assessment server **118** can compare the first risk assessment data to the second risk assessment data, and, if the first assessment data is the same as or similar to the second risk assessment data, the risk assessment server **118** may validate the first risk assessment data. The risk assessment server **118** can use other suitable techniques for validating the first risk assessment data and the second risk assessment data. Additionally, by validating risk assessment data from different sources, the risk assessment server **118**, the risk assessment computing system **130**, or the like can use less memory and can use memory more efficiently for validating the risk assessment data than a set of computing systems performing separate validation processes.

(36) At block **210**, the process **200** involves outputting the validated risk assessment data for display on the user interface. The risk assessment server **118** can transmit the validated first risk assessment data and the second risk assessment data to the user interface for display to the first entity or any other suitable entities. The validated risk assessment data can include a set of risk assessment indicators such as the first risk assessment indicator, etc. Each risk assessment indicator can be displayed on a separate user interface element. For example, the validated first risk assessment indicator can be displayed on the first user interface element, and so on. The first entity can view the risk assessment indicators and can decide whether to proceed with an interaction such as requesting online computing resources, requesting a loan, submitting the URLA, etc.

(37) At block **212**, the process **200** involves initiating an interaction based on the validated risk assessment data. In response to the first entity determining to continue with the interaction, the risk assessment server **118** can initiate the interaction using the validated risk assessment data. For example, the risk assessment server **118** can pre-fill an application with the validated risk

assessment data and can transmit the application to a separate computing system such as the client computing system **104**, etc. The separate computing system can make decisions about the first entity using the validated risk assessment data. Accordingly, by consolidating the data sources and validating the risk assessment data before initiating the interaction, the risk assessment computing system **130** can reduce an amount of computer memory used and can increase a processing speed of the separate computing system with respect to decision-making about the first entity.

Additionally, the security of the validation process can be improved since the risk assessment data is consolidated and is transmitted, or otherwise handled, less often than other validation processes. (38) In one example, the first user interface element can correspond to a credit score or other risk score, the second user interface element can correspond to a debt-to-income ratio, and the third user interface element can correspond to assets. The entity can interact with the first user interface element to provide information usable to determine the credit score or other risk score. For example, the first entity can input identifying information, such as name, social security number, etc., into the first subsequent user interface, and the risk assessment server **118** can use the pre-existing risk assessment data **124** or other suitable information to determine the risk assessment indicator corresponding to the credit score or other risk score of the first entity. The first entity can interact with the second user interface element for inputting income and debt information. For example, a second subsequent user interface, provided in response to the first entity interacting with the second user interface element, can be configured to receive and transmit employment information, income information, and debt information to the risk assessment server **118**, which can use the pre-existing risk assessment data **124** to validate and output the risk assessment indicator corresponding to the debt-to-income ratio.

(39) The first entity can interact with the third user interface element for inputting asset information. For example, a third subsequent user interface, provided in response to the first entity interacting with the third user interface element, can be configured to receive and transmit the asset information to the risk assessment server **118**. In some examples, the third subsequent user interface can allow the first entity to login to an external resource **107** for retrieving the second risk assessment data. Upon the first entity logging into the external resource **107**, a token can be generated and used by the risk assessment server **118**, or other suitable computing system, to retrieve the second risk assessment data from the second entity. The risk assessment server **118** can compare the second risk assessment data to the first risk assessment data for validating the first risk assessment data, and, in some examples, the risk assessment server **118** can use the pre-existing risk assessment data **124** to validate the second risk assessment data. The second risk assessment data can be used to perform various other suitable operation.

(40) Once validated, the first risk assessment data and the second risk assessment data can be output by the risk assessment server **118**. For example, the risk assessment server **118** can output the risk assessment indicators of the first entity for display on the user interface. The first entity, or other suitable entity, can view the risk assessment indicators for determining whether to proceed with an interaction. In some examples, the first entity may decide to proceed with an interaction, and, in response to this determination, the risk assessment server **118** may transmit the validated risk assessment data, and other suitable data, to a client computing system **104** for initiating the interaction.

(41) In some examples, the risk assessment server **118** can transmit a responsive message including at least the validated first risk assessment data and the validated second risk assessment data for use in controlling access of the first entity to one or more interactive computing environments. The risk assessment server **118** can transmit the responsive message to a remote computing system such as the client computing system **104**, and the like. The responsive message may cause computing resources to be allocated, or to not be allocated, to the first entity based on the validated first risk assessment data and the validated second risk assessment data. Additionally or alternatively, the responsive message may cause the interaction to be initiated.

(42) In some examples, the risk assessment server **118** can also generate reports in required formats based on the data received from the various sources. For instance, a look-up table can be built to map the data from the sources to the corresponding fields of the reports. The risk assessment server **118** can convert the data into the required format before filling in the report. Alternatively, or additionally, a machine learning model based on document analysis techniques and/or natural language processing techniques can be built and trained to identify or categorize the data received from the sources into the types of data required by the fields in the reports. Based on the output of the machine learning model, the risk assessment server **118** can fill in the report with the identified data to generate the reports. In some examples, the machine learning model is a neural-network-based model, such as the bidirectional encoder representations from transformers (BERT) model.

(43) Example of a Data Flow of Computing Systems Described Herein

(44) FIG. **3** is a flow diagram depicting a flow of risk assessment data in a network environment **300** for consolidating data sources to expedite validation of the risk assessment data according to certain aspects of the present disclosure. The network environment **300** can include the risk assessment computing system **130** and other suitable computing devices or systems. The risk assessment computing system **130** can receive pre-existing risk assessment data **124**, which can include historical risk scores, employment information, income information, historical credit scores, and other suitable pre-existing risk assessment data. In some examples, the pre-existing risk assessment data **124** can be received by the risk assessment computing system **130** from third-party or other suitable exterior data sources.

(45) The risk assessment computing system **130** can provide and can communicate with a risk assessment interactive computing environment **302**. The computing environment **302** can include a user interface **304** for risk assessment, a set of subsequent user interfaces **306** for risk assessment, an API **308** for risk assessment, and other suitable components. The user interface **304** can be provided to one or more entities **310** that can provide input into the user interface **304** for accessing the subsequent user interfaces **306**. The subsequent user interfaces **306** can be provided to the entities **310** for receiving risk assessment data, and other suitable data, from the entities **310** and for risk assessment.

(46) The API **308** can make API calls between the risk assessment interactive computing environment **302** and the risk assessment computing system **130**. The API **308** can make API calls between the risk assessment interactive computing environment **302** and other suitable entities. For example, the API **308** can make API calls between the computing environment **302** and exterior data and verifications **312**. The exterior data and verifications **312** can be included on a computing device or system of a separate entity, such as a financial institution or other service-providing-entity, and can include validation data, additional risk assessment data, and other suitable data about the first entity. The API **308** can make an API call to the exterior data and verifications **312** to cause the computing environment **302** to receive the validation data and additional risk assessment data about the first entity. The API **308** can make an API call to the risk assessment computing system **130** to transmit the risk assessment data received from the entities **310** and from the exterior data and verifications **312** to the risk assessment computing system **130**.

(47) The risk assessment computing system **130** can receive, at a data ingestion **314** and from the API **308**, the pre-existing risk assessment data **124**, the risk assessment data from the entities **310**, and the exterior data and verifications **312** from the separate entity. For example, the risk assessment computing system **130** can provide the risk assessment mobile application **114** or the risk assessment web application **115** to one or more of the entities **310**. The data ingestion **314** can receive the data, for example from the risk assessment mobile application **114** or the risk assessment web application **115**, can transmit the data within the risk assessment computing system **130** for processing, and can perform other suitable tasks relating to the received data. For example, the data ingestion **314** can transmit the received data to a data parser **316** that can parse or otherwise process the received data. The data parser **316** can receive the risk assessment data, for

example from the data ingestion **314** and via the risk assessment mobile application **114** or the risk assessment web application **115**, and can determine risk assessment indicators. For example, the data parser **316** can receive debt information, employment information, and income information about the first entity and can determine a debt-to-income ratio risk indicator for the first entity. The data parser **316** can determine the remaining risk indicators and can output risk assessment indicators **318**. In some examples, one or more machine-learning models, for example a document-reading artificial intelligence or other suitable machine-learning model, can be applied to receive, for example via the risk assessment mobile application **114** or the risk assessment web application **115**, the risk assessment data and extract the risk assessment indicators **318**.

(48) The risk assessment indicators **318** can include various risk indicators. For example, the risk assessment indicators **318** can include a risk score for the first entity, a credit score for the first entity, a debt-to-income ratio of the entity, assets required for the entity, and other suitable risk indicators relating to the entity. In some examples, the risk assessment indicators **318** can additionally include data from the exterior data and verifications **312** such as a parsed asset report from a service-providing entity, transcripts from a government institution, data relating to account integrations, and other suitable data from the exterior data and verifications **312**. The risk assessment indicators **318** can be transmitted, for example via one or more API calls, to the risk assessment interactive computing environment **302** for display on the user interface **304**. For example, each risk indicator of the risk assessment indicators **318** can be displayed on a corresponding user interface element of the user interface **304**. The risk assessment indicators **318** can be transmitted and displayed in other suitable manners for informing the entities **310** of the risk assessment indicators **318**.

(49) Examples of a User Interface

(50) FIG. 4 is an example of a user interface **400** that can be used to consolidate data sources for expediting validation of risk assessment data according to certain aspects of the present disclosure. The user interface **400** can include a set of user interface elements **402a-d**. While four user interface elements **402a-d** are illustrated on the user interface **400**, other suitable amounts (e.g., less than four or more than four) of user interface elements **402** can be included on the user interface **400**.

(51) The user interface element **402a** can include a title **404** and one or more input fields **406**. As illustrated, the title **404** is “GOALS” and the input fields **406** relate to the goals such as a type of product, an amount, etc. The user interface element **402a** can be used by the first entity to adjust the risk assessment associated with obtaining a mortgage. For example, adjusting information in the user interface element **402a** can affect whether the first entity is approved or denied for a requested product or service based on the risk assessment of the first entity. The user interface element **402a** can include other suitable titles and input fields **406** and can be used for any other suitable purpose with respect to the first entity.

(52) The user interface element **402b** can include a risk indicator value **408a**, a risk indicator measure **410a**, and an interactive button **412**. The risk indicator value **408a** can be or otherwise include a numeric or other suitable type of value that can represent one aspect of risk assessment with respect to the first entity. For example, the risk indicator value **408a** is illustrated as a credit score, which can be used to apply for the mortgage, of the first entity, but other suitable risk indicators can be displayed as the risk indicator value **408a**. The risk indicator measure **410a** can be a visual representation of how good the risk indicator value **408a** is. For example, a high risk score may be displayed as an arrow in the green area of the risk indicator measure **410a** while a low risk score may be displayed as the arrow in the red area of the risk indicator measure **410a**. Other suitable representations of the risk indicator measure **410a** can be used.

(53) The first entity or other suitable entity can interact with (e.g., click) the interactive button **412**. Interacting with the interactive button **412** can cause the user interface **400** to present or otherwise provide a subsequent user interface for the first entity. The subsequent user interface can be used to

input information relating to the risk indicator associated with the user interface element **402b**. For example, the first entity can input information relating to the risk score in the subsequent user interface.

(54) The user interface element **402c** can include a risk indicator value **408b**, a risk indicator measure **410b**, and the interactive button **412**. The risk indicator value **408b** can be or otherwise include a numeric or other suitable type of value that can represent one aspect of risk assessment with respect to the first entity. For example, the risk indicator value **408b** is illustrated as a debt-to-income ratio, which can indicate a maximum mortgage amount, of the first entity, but other suitable risk indicators can be displayed as the risk indicator value **408b**. The risk indicator measure **410b** can be a visual representation of how good the risk indicator value **408b** is. For example, a low debt-to-income ratio may be displayed as an arrow in the green area of the risk indicator measure **410b** while a high debt-to-income ratio may be displayed as the arrow in the red area of the risk indicator measure **410b**. Other suitable representations of the risk indicator measure **410b** can be used.

(55) The first entity can interact with (e.g., click) the interactive button **412**. Interacting with the interactive button **412** can cause the user interface **400** to present or otherwise provide a subsequent user interface for the first entity. The subsequent user interface can be used to input information relating to the risk indicator associated with the user interface element **402c**. For example, the first entity can input information relating to debt information, income information, employment information, or other suitable information in the subsequent user interface.

(56) The user interface element **402d** can include a risk indicator value **408c**, a risk indicator measure **410c**, and an interactive button **412**. The risk indicator value **408c** can be or otherwise include a numeric or other suitable type of value that can represent one aspect of risk assessment with respect to the first entity. For example, the risk indicator value **408c** is illustrated as assets controlled by the first entity, but other suitable risk indicators can be displayed as the risk indicator value **408c**. The assets controlled by the first entity can indicate, for example, an initial resource allocation for which the first entity can qualify with respect to a mortgage. The risk indicator measure **410c** can be a visual representation of how good the risk indicator value **408c** is. For example, a large amount of resources of the first entity may be displayed as an arrow in the green area of the risk indicator measure **410c** while a small amount of resources of the first entity may be displayed as the arrow in the red area of the risk indicator measure **410c**. Other suitable representations of the risk indicator measure **410c** can be used.

(57) The first entity can interact with (e.g., click) the interactive button **412**. Interacting with the interactive button **412** can cause the user interface **400** to present or otherwise provide a subsequent user interface for the first entity. The subsequent user interface can be used to input information relating to the risk indicator associated with the user interface element **402d**. For example, the first entity can input information relating to the assets controlled by the first entity in the subsequent user interface. Additionally, the subsequent user interface can provide an option to login to a separate computing system to retrieve additional risk assessment data for the first entity.

(58) Each of the user interface elements **402b-d** can additionally include an option to access additional information. For example, the option can be a link, a button, or other suitable interactive field with which the first entity can interact. The option can display information to the first entity, and the information can be related to the associated risk assessment indicator. For example, interacting with the option on the user interface element **402b** may provide more information relating to the risk score, for example what it means, how to improve the risk score, etc. Interacting with the option on the other user interface elements **402c-d** can provide similar information about the debt-to-income ratio and the assets risk indicators.

(59) The user interface **400** can additionally include a display **414** that can indicate whether the risk assessment of the first entity is adequately complete. For example, as illustrated on the user interface **400**, the display **414** indicates that the first entity is ready to submit the risk assessment

indicators for requesting a product or service. Alternatively, the display **414** can display other suitable values such as “NOT READY” or other suitable display values. In some examples, the display **414** can be interactive. For example, the consumer can interact with the display **414**, when prompted, to submit the risk assessment data.

(60) FIG. 5 is an example of a subsequent user interface **500** that can be used to receive risk assessment data to consolidate data sources for expediting validation of risk assessment data according to certain aspects of the present disclosure. The subsequent user interface **500** can include various fields and information for guiding the first entity to provide accurate and complete information for determining risk indicators. In some examples, the subsequent user interface **500**, or any derivative thereof, can be accessed in response to interacting with the interactive button **412** of the user interface **400**.

(61) As illustrated, the subsequent user interface **500** is an example of a user interface for gathering information relating to a risk score of the first entity. The subsequent user interface **500** can include a title **502**, a name field **504**, identifying information fields **506**, an authorization field **508**, and a submission button **510**. The title **502** can display information indicating a type of information contained within the subsequent user interface **500** to a user of the subsequent user interface **500**. For example, as illustrated, the title **502** is “CREDIT INTERVIEW,” which indicates that the subsequent user interface **500** relates to credit score information.

(62) The name field **504** can be completed by the first entity. For example, the first entity can input their name into the name field **504** for allowing the risk assessment computing system **130** to determine a risk score of the first entity. Additionally, the first entity can input identifying information into the identifying information fields **506** for allowing the risk assessment computing system **130** to determine the risk score. The identifying information fields **506** can include a social security number, a date of birth, an address, and other suitable fields for receiving identifying information.

(63) Upon completion of the subsequent user interface **500**, the first entity may provide an authorization via the subsequent user interface **500**. For example, the first entity can interact with the authorization field **508** for indicating that the first entity believes the information input into the subsequent user interface **500** is accurate and complete. In some examples, the authorization indicator can include a simple checkbox, a signature field, or other suitable authorization indicators. In response to providing the authorization, the first entity may interact with the submission button **510**. By interacting with the submission button **510**, the first entity may cause the information input into the subsequent user interface **500** to be transmitted to the risk assessment computing system **130** for determining corresponding risk assessment indicators.

(64) While the subsequent user interface **500** is illustrated for a credit interview for determining a credit score, other suitable risk indicators can be determined using information from various different subsequent user interfaces. For example, the subsequent user interface **500**, instead of a credit interview, can include an employment, income, or debt interview, an asset interview, an identity interview, or other suitable types of subsequent user interfaces **500**. Each risk indicator used or otherwise determined by the risk assessment computing system **130** can have a corresponding subsequent user interface **500** for allowing the first entity to provide associated risk assessment data.

(65) Example of Computing System

(66) Any suitable computing system or group of computing systems can be used to perform the operations for the risk assessment operations described herein. For example, FIG. 6 is a block diagram depicting an example of a computing device **600**, which can be used to implement the risk assessment server **118** or other suitable components of the operating environment **100**. The computing device **600** can include various devices for communicating with other devices in the operating environment **100**, as described with respect to FIG. 1. The computing device **600** can include various devices for performing one or more data consolidation operations, data validation

operations, or other suitable operations described above with respect to FIGS. 1-5.

(67) The computing device **600** can include a processor **602** that is communicatively coupled to a memory **604**. The processor **602** can execute computer-executable program code stored in the memory **604**, can access information stored in the memory **604**, or a combination thereof. Program code may include machine-executable instructions that may represent a procedure, a function, a subprogram, a program, a routine, a subroutine, a module, a software package, a class, or any combination of instructions, data structures, or program statements. A code segment may be coupled to another code segment or a hardware circuit by passing or receiving information, data, arguments, parameters, or memory contents. Information, arguments, parameters, data, etc., may be passed, forwarded, or transmitted via any suitable means including memory sharing, message passing, token passing, network transmission, among others.

(68) Examples of a processor **602** can include a microprocessor, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or any other suitable processing device. The processor **602** can include any suitable number of processing devices, including one. The processor **602** can include or communicate with a memory **604**. The memory **604** can store program code that, when executed by the processor **602**, causes the processor **602** to perform the operations described herein.

(69) The memory **604** can include any suitable non-transitory computer-readable medium. The computer-readable medium can include any electronic, optical, magnetic, or other storage device capable of providing a processor with computer-readable program code or other program code. Non-limiting examples of a computer-readable medium can include a magnetic disk, memory chip, optical storage, flash memory, storage class memory, ROM, RAM, an ASIC, magnetic storage, or any other medium from which a computer processor can read and execute program code. The program code may include processor-specific program code generated by a compiler or an interpreter from code written in any suitable computer-programming language. Examples of suitable programming language can include Hadoop, C, C++, C #, Visual Basic, Java, Python, Perl, JavaScript, ActionScript, etc.

(70) The computing device **600** may also include a number of external or internal devices such as input or output devices. For example, the computing device **600** is illustrated with an input/output interface **608** that can receive input from input devices or provide output to output devices. A bus **606** can also be included in the computing device **600**. The bus **606** can communicatively couple one or more components of the computing device **600**.

(71) The computing device **600** can execute program code **614** that can include the risk assessment mobile application **114**. The program code **614** for the risk assessment mobile application **114** may be resident in any suitable computer-readable medium and may be executed on any suitable processing device. For example, as depicted in FIG. 6, the program code **614** for the risk assessment mobile application **114** can reside in the memory **604** at the computing device **600** along with the program data **616** associated with the program code **614**, such as the pre-existing risk assessment data **124**. Executing the risk assessment mobile application **114** can configure the processor **602** to perform the operations described herein.

(72) In some aspects, the computing device **600** can include one or more output devices. One example of an output device can be the network interface device **610** depicted in FIG. 6. A network interface device **610** can include any device or group of devices suitable for establishing a wired or wireless data connection to one or more data networks described herein. Non-limiting examples of the network interface device **610** can include an Ethernet network adapter, a modem, etc.

(73) Another example of an output device can include the presentation device **612** depicted in FIG. 6. A presentation device **612** can include any device or group of devices suitable for providing visual, auditory, or other suitable sensory output. Non-limiting examples of the presentation device **612** can include a touchscreen, a monitor, a speaker, a separate mobile computing device, etc. In some aspects, the presentation device **612** can include a remote client-computing device that

communicates with the computing device **600** using one or more data networks described herein. In other aspects, the presentation device **612** can be omitted.

(74) The foregoing description of some examples has been presented only for the purpose of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Numerous modifications and adaptations thereof will be apparent to those skilled in the art without departing from the spirit and scope of the disclosure.

Claims

1. A method that includes one or more processing devices performing operations comprising: providing, from a risk assessment computing system, a user interface to a first entity, the user interface comprising a plurality of user interface elements corresponding to a plurality of risk assessment indicators, the plurality of user interface elements configured to receive first risk assessment data corresponding to at least a subset of the plurality of risk assessment indicators; receiving, via the user interface, and at the risk assessment computing system, the first risk assessment data that is associated with the first entity; generating a digital token in response to the first entity performing an interaction via the user interface; using the digital token to access second risk assessment data that is associated with the first entity, wherein the second risk assessment data is provided to the risk assessment computing system by a second, digitally secured entity; consolidating, at the risk assessment computing system, the first risk assessment data and the second risk assessment data; validating, by the risk assessment computing system and during the same validation process, the first risk assessment data and the second risk assessment data, wherein the validation process comprises: comparing the first risk assessment data received via the user interface and the second risk assessment data received via the second, digitally secured entity based on predefined validation rules specific to a type and source of the respective data; identifying, based on the comparison, one or more discrepancies between the first risk assessment data and the second risk assessment data; and resolving the one or more discrepancies between the first risk assessment data and the second risk assessment data to generate a validated first risk assessment data and a validated second risk assessment data; outputting the validated first risk assessment data and the validated second risk assessment data for display on the user interface; transmitting, to a remote computing device, a responsive message including at least the validated first risk assessment data and the validated second risk assessment data; and controlling access of the first entity to one or more interactive computing environments using at least the responsive message.
2. The method of claim 1, wherein the interaction performed by the first entity comprises providing login information to the user interface.
3. The method of claim 1, wherein receiving the first risk assessment data via the user interface comprises receiving input from the first entity for causing a subsequent user interface to be provided to the first entity for receiving additional first risk assessment data.
4. The method of claim 3, wherein the subsequent user interface comprises a subsequent plurality of user interface elements usable by the first entity to cause the additional first risk assessment data to be received.
5. The method of claim 1, wherein transmitting the responsive message comprises initiating an interaction using the validated first risk assessment data and the validated second risk assessment data, wherein the interaction includes a request by the first entity for access to one or more computing resources.
6. The method of claim 1, wherein validating the first risk assessment data and the second risk assessment data comprises: determining that the first risk assessment data matches the second risk assessment data; and determining, based on the first risk assessment data matching the second risk assessment data, that the first risk assessment data and the second risk assessment data are valid.
7. The method of claim 1, wherein validating the first risk assessment data and the second risk

assessment data comprises: determining that the first risk assessment data and the second risk assessment data match third-party risk assessment data; and determining, based on the first risk assessment data and the second risk assessment data matching the third-party risk assessment data, that the first risk assessment data and the second risk assessment data are valid.

8. A system comprising: a processor; and a non-transitory computer-readable medium comprising instructions that are executable by the processor to cause the processor to perform operations comprising: providing a user interface to a first entity, the user interface comprising a plurality of user interface elements corresponding to a plurality of risk assessment indicators; receiving, via the user interface, first risk assessment data that is associated with the first entity; generating a digital token in response to the first entity performing an interaction via the user interface; using the digital token to access second risk assessment data that is associated with the first entity, wherein the second risk assessment data is provided to the risk assessment computing system by a second, digitally secured entity; validating the first risk assessment data and the second risk assessment data during the same validation process, wherein the validation process comprises: comparing the first risk assessment data received via the user interface and the second risk assessment data received via the second, digitally secured entity based on predefined validation rules specific to a type and source of the respective data; identifying, based on the comparison, one or more discrepancies between the first risk assessment data and the second risk assessment data; and resolving the one or more discrepancies between the first risk assessment data and the second risk assessment data to generate a validated first risk assessment data and a validated second risk assessment data; outputting the first risk assessment data and the second risk assessment data for display on the user interface; and transmitting, to a remote computing device, a responsive message including at least the validated first risk assessment data and the validated second risk assessment data; and controlling access of the first entity to one or more interactive computing environments using at least the responsive message.

9. The system of claim 8, wherein the interaction performed by the first entity comprises providing login information to the user interface.

10. The system of claim 8, wherein the operation of receiving the first risk assessment data via the user interface comprises receiving input from the first entity for causing a subsequent user interface to be provided to the first entity for receiving additional first risk assessment data.

11. The system of claim 10, wherein the subsequent user interface comprises a subsequent plurality of user interface elements usable by the first entity to cause the additional first risk assessment data to be received.

12. The system of claim 8, wherein the operation of transmitting the responsive message comprises initiating an interaction using the validated first risk assessment data and the validated second risk assessment data, wherein the interaction includes a request by the first entity for access to one or more computing resources.

13. The system of claim 8, wherein the operation of validating the first risk assessment data and the second risk assessment data comprises: determining that the first risk assessment data matches the second risk assessment data; and determining, based on the first risk assessment data matching the second risk assessment data, that the first risk assessment data and the second risk assessment data are valid.

14. The system of claim 8, wherein the operation of validating the first risk assessment data and the second risk assessment data comprises: determining that the first risk assessment data and the second risk assessment data match third-party risk assessment data; and determining, based on the first risk assessment data and the second risk assessment data matching the third-party risk assessment data, that the first risk assessment data and the second risk assessment data are valid.

15. A non-transitory computer-readable medium comprising instructions that are executable by a processing device for causing the processing device to perform operations comprising: providing a user interface to a first entity, the user interface comprising a plurality of user interface elements

corresponding to a plurality of risk assessment indicators; receiving, via the user interface, first risk assessment data that is associated with the first entity; generating a digital token in response to the first entity performing an interaction via the user interface; using the digital token to access second risk assessment data that is associated with the first entity, wherein the second risk assessment data is provided to the risk assessment computing system by a second, digitally secured entity; validating the first risk assessment data and the second risk assessment data during the same validation process, wherein the validation process includes: comparing the first risk assessment data received via the user interface and the second risk assessment data received via the second, digitally secured entity based on predefined validation rules specific to a type and source of the respective data; identifying, based on the comparison, one or more discrepancies between the first risk assessment data and the second risk assessment data; and resolving the one or more discrepancies between the first risk assessment data and the second risk assessment data to generate a validated first risk assessment data and a validated second risk assessment data; outputting the first risk assessment data and the second risk assessment data for display on the user interface; transmitting, to a remote computing device, a responsive message including at least the validated first risk assessment data and the validated second risk assessment data; and controlling access of the first entity to one or more interactive computing environments using at least the responsive message.

16. The non-transitory computer-readable medium of claim 15, wherein interaction performed by the first entity comprises providing login information to the user interface.

17. The non-transitory computer-readable medium of claim 15, wherein the operation of receiving the first risk assessment data via the user interface comprises receiving input from the first entity for causing a subsequent user interface to be provided to the first entity, and wherein the subsequent user interface comprises a subsequent plurality of user interface elements usable by the first entity to cause the first risk assessment data to be received.

18. The non-transitory computer-readable medium of claim 15, wherein the operation of transmitting the responsive message comprises initiating an interaction using the validated first risk assessment data and the validated second risk assessment data, wherein the interaction includes a request by the first entity for access to one or more computing resources.

19. The non-transitory computer-readable medium of claim 15, wherein the operation of validating the first risk assessment data and the second risk assessment data comprises: determining that the first risk assessment data matches the second risk assessment data; and determining, based on the first risk assessment data matching the second risk assessment data, that the first risk assessment data and the second risk assessment data are valid.

20. The non-transitory computer-readable medium of claim 15, wherein the operation of validating the first risk assessment data and the second risk assessment data comprises: determining that the first risk assessment data and the second risk assessment data match third-party risk assessment data; and determining, based on the first risk assessment data and the second risk assessment data matching the third-party risk assessment data, that the first risk assessment data and the second risk assessment data are valid.
